Tomography-Based 3-D Anisotropic Elastography Using Boundary Measurements

Yi Liu, L. Z. Sun*, and Ge Wang, *Fellow, IEEE*

***Abstract*—While ultrasound- and magnetic resonance-based elastography techniques have proved to be powerful biomedical imaging tools, most approaches assume isotropic material properties. In this paper, a general framework is developed for tomography-based anisotropic elastography. An anatomically well-motivated piece-wise homogeneous model is proposed to represent a class of biological objects consisting of different regions. With established tomography modality, static displacements are measured on the entire external and internal boundaries, and the force distribution is recorded on part of the external surface. A principle is proposed to identify the anisotropic elastic moduli of the constituent regions with the obtained boundary measurements. The reconstruction procedure is optimization-based with minimizing an objective function that measures the difference between the predicted and observed displacements. Analytic gradients of the objective function with respect to the elastic moduli are calculated using an adjoint method, and are utilized to significantly improve the numerical efficiency. Simulations are performed to identify the elastic moduli in a breast phantom consisting of soft tissue and a hard tumor. For isotropic phantom, one set of the boundary measurements enables unique reconstruction results for the tissue and tumor. For anisotropic phantom, however, multiple sets of the measurements corresponding to different deformation modes become necessary.**

***Index Terms*—Adjoint method, anisotropic biomaterial, breast imaging, elastography, finite-element method, tomography.**

I. Introduction

PATHOLOGICAL changes often alter the elastic properties of tissues. For instance, a hard lump felt in a breast often indicates a tumor [1], because breast carcinoma is significantly stiffer than the surrounding tissues [2], [3]. Elastography is correspondingly developed to quantify and visualize distribution of the elastic modulus (stiffness) in biological tissues by taking mechanical and biomedical imaging measurements as inputs. The outputs of elastography are used to assist diagnosis of tumors in the breast, prostate, and liver, among other organs, by recognizing the difference in stiffness between healthy and abnormal tissues. Compared to qualitative palpation, elastography may have much higher sensitivity and specificity, and be more effective in detecting deeply embedded small tumors.

Most elastography models take a dynamic or quasi-static displacement field as input for identification of the elastic parameters in an object. The displacement field is typically measured with ultrasound (US) or magnetic resonance (MR) imaging techniques [4]–[16]. Muthupillai *et al.* [6], Manduca *et al.* [7] and McCracken *et al.* [8] reconstructed the shear modulus distribution from MR measurement of the acoustic wave. Plewes *et al.* [9], Samani *et al.* [10] and Weaver *et al.* [11] conducted finite element computations for the elastic modulus distribution in a soft tissue containing a suspicious tumor, based on MR measurement of the deformation. Kallel and Bertrand [12], Ophir *et al.* [13], [14], Skovoroda *et al.* [15] and Taylor *et al.* [16] developed computational methodologies for inhomogeneous elastography using the displacement field measured with US techniques. Oberai *et al.* [17] recently adopted an adjoint method and proposed an efficient numerical scheme for reconstructing the nonuniform shear modulus in incompressible isotropic materials. There are also elastography reconstructions with use of boundary measurements; for instance, the finite difference method of Cox and Gockenbach [18] for the Lamé parameters in two–dimensional isotropic material, a finite element approach of Zhu *et al.* [19] for inhomogeneous Young's modulus, and semi-analytic methods of Kang *et al.* [20] and Liu *et al.* [21] to identify tumor-like inclusions in isotropic media.

While the above studies provide new insight into detecting tumors and other pathological abnormalities, they assume isotropic elastic behaviors for biotissues. It has been recognized that certain soft tissues, like muscles and glands, are anisotropic with respect to elastic deformation [22]. Also, observations indicate that breast tumors tend to be anisotropic [23], [24]. Gennisson *et al.* [25] calculated distribution of transversely isotropic elastic parameters using characteristic wavelength measured with an ultrasound polarization technique. Sinkus *et al.* [26] recently derived formulas for MR-based elastography reconstruction, assuming transverse isotropy for the tissues, and conducted inhomogeneous phantom and *ex vivo* simulations. These anisotropic elastography studies demonstrate the need for generalized formulas and extra measurements, compared with isotropic cases. Therefore, consideration of elastic anisotropy is essential for clinical elastography applications.

This paper aims to develop a tomography (e.g., CT)-based, three-dimensional (3-D) static anisotropic elastography. A piece-wise homogeneous, anisotropic elastic model is proposed to simulate tumor-containing objects, as described in Section II-A. The measurements employed in the elastography reconstruction are detailed in Section II-B, which include

Manuscript received April 12, 2005; revised July 29, 2005. This work was supported in part by the National Institutes of Health (NIH) under NIBIB Grant EB002667-01, in part by the University of Iowa under an Informatics Initiative Grant, and in part by the US Army's Breast Cancer Research Program Concept Award. The Associate Editor responsible for coordinating the review of this paper and recommending its publication was M. Fatemi. *Asterisk indicates corresponding author.*

Y. Liu and G. Wang are with the Department of Radiology, University of Iowa Hospitals and Clinics, Iowa City, IA 52242 USA.

*L. Z. Sun is with the Department of Civil and Environmental Engineering, The University of Iowa, 4114 Seamans Center, Iowa City, IA 52242-1527 USA (e-mail: lizhi-sun@uiowa.edu).

Corresponding author: L. Z. Sun, lizhi-sun@uiowa.edu

Digital Object Identifier 10.1109/TMI.2005.857232

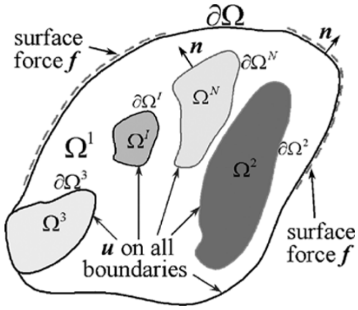

Fig. 1. A piece-wise homogeneous anisotropic elastic medium consisting of N regions. Displacements $\boldsymbol{u}$ are measured on external surface $\partial\Omega$ and interfaces $\partial\Omega^I$, and distributed surface forces $\boldsymbol{f}$ are measured on part of $\partial\Omega$.

static-displacement on the entire external and internal boundaries, and surface force distribution on a portion of the external boundary. A principle is then proposed in Section II-C for identification of the elastic moduli of the described anisotropic media. A numerical scheme for the reconstruction is presented in Section III. It is optimization-based and implements user-supplied analytic gradients of the objective function, which significantly enhance the numerical efficiency. Effort for analytically computing the gradients is shown to be minimal with use of an *adjoint* displacement field. The scheme is also capable of handling multiple sets of measurements as needed for unique identifiability. Numerical simulations are presented in Section IV to identify the elastic moduli in a 3-D breast phantom with an embedded hard tumor. These calculations demonstrate the feasibility of the proposed method for anisotropic elastography using boundary measurements.

II. Geometric Model and Principle

A. Piece-Wise Homogeneous Elastic Media

In this paper, a tumor-containing biomedical object is modeled as an elastic medium Ω consisting of N homogeneous (uniform) linear anisotropic regions. As shown schematically in Fig. 1, regions Ω^I $(I = 2, 3, \ldots, N)$ are embedded fully or partly into the matrix Ω^1. The elastic properties in a region Ω^I are, thus, assumed uniform, and are described with an anisotropic elastic modulus $\boldsymbol{C}^I$. The external surface of the medium $\partial\Omega$ includes $\partial\Omega_u$ where the displacements $\boldsymbol{u}$ are prescribed in experiments, and $\partial\Omega_t$ where distributed surface forces $\boldsymbol{f}$ are applied. The boundary of region Ω^I $(I = 2, 3, \ldots, N)$ is called an internal boundary, and is denoted as $\partial\Omega^I$. Such a piece-wise homogeneous model is anatomically well-motivated for some organs that contain suspicious tumors, and has been adopted in finite-element simulations [27] and elastography studies [9]. In a breast, for instance, the fatty/connective tissue is modeled as the matrix Ω^1, and the glands and tumors (if exist) are modeled as the embedded Ω^I $(I = 2, 3, \ldots, N)$, respectively. The mechanical properties in a real tissue are not exactly uniform; however, their variations should be relatively small compared with the difference between the constituent regions, and the characteristic length for material heterogeneity is typically shorter than the size of the anatomic structures. Therefore, most tissues can be effectively considered homogeneous in mechanical properties. For the same reasons, and because tumors are much stiffer than the neighboring tissues, tumors can also be modeled as effectively homogeneous. In clinical practice, the anatomic structures in an object, and the external and internal boundaries, are three-dimensionally depicted using established CT modality. In a case where the internal boundaries are fuzzy in the tomographic images, definition of the anatomic structures may be subjective and prone to human error. Various biomedical-imaging techniques can be implemented to enhance the contrast on the interfaces and improve the boundary detection. On the other hand, the effects of geometric infidelity on elastography reconstructions need to be explored, as reported in our recent work [28].

Linear elastic deformation is assumed in this paper. At a material point $\boldsymbol{x}$, the strain $\boldsymbol{\varepsilon}(\boldsymbol{x})$ linearly depends on the displacement $\boldsymbol{u}(\boldsymbol{x})$, and the stress $\boldsymbol{\sigma}(\boldsymbol{x})$ linearly depends on the strain, i.e.,

$$\boldsymbol{\varepsilon}(\boldsymbol{x}) = \boldsymbol{\varepsilon}(\boldsymbol{u}) = \frac{1}{2}\left(\nabla \boldsymbol{u}(\boldsymbol{x}) + (\nabla \boldsymbol{u}(\boldsymbol{x}))^T\right) \text{ and } \boldsymbol{\sigma}(\boldsymbol{x}) = \boldsymbol{C}(\boldsymbol{x}) : \boldsymbol{\varepsilon}(\boldsymbol{u}). \tag{1}$$

The stiffness or elastic modulus $\boldsymbol{C}$ is a symmetric, positive definite fourth-order tensor. While it is spatially dependent in general, it is assumed to be piece-wise uniform for the problems of interest, i.e., $\boldsymbol{C}(\boldsymbol{x}) = \boldsymbol{C}^I$ when $\boldsymbol{x} \in \Omega^I$. It is noted that the linear elastic relation in (1) demonstrates all deformations are small which serves as a first-order engineering approximation. It is, therefore, reasonable to model the displacement field as being continuous, at least in an initial formulation. It is acknowledged that the displacement discontinuity may occur in clinical practice. One feasible way to approximate the discontinuity is to introduce a thin layer of soft material on the boundary of the mobile region of interest. Accurate modeling of such discontinuity needs large-deformation descriptions, which is considered beyond the scope of this paper.

B. Boundary Measurements

For the piece-wise homogeneous elastic media, static displacements $\boldsymbol{u}$ are assumed to be measured on the *entire* outer surface $\partial\Omega$ and *all* the internal boundaries $\partial\Omega^I$ $(I = 2, 3, \ldots, N)$, by tracking landmarks on the external and internal boundaries during the deformation, using well-developed elastic image registration techniques [29]. As a result, both normal and tangential displacements may be estimated reliably based on the point-by-point correspondence established in the registration process. Because these measurements are static and surface type, they may not carry the same amount of information about the material properties as the full-field, dynamic/quasistatic displacements commonly used in the US or MR elastography studies. However, with the CT modality, higher spatial resolution and geometric fidelity may be obtained than with US and MR, and the anatomic structures can be more accurately identified. Furthermore, tomography-based elastography does not require dynamic mechanical loadings, and has the potential of being easily applied in some clinical settings. It is noted that the higher the spatial resolution with a CT scanner, the more detailed structures it can reveal. Hence, generally speaking, the higher accuracy may be achieved in displacement measurements. A systematic evaluation of the relationship among all the relevant information content and density of the images is important and will be studied in the future. Distributed surface forces $\boldsymbol{f}$ are essential for this type of elastography reconstruction. In typical clinical applications, $\boldsymbol{f}$ can be measured with micro-electro-mechanical sensors on part of the surface, where compression or other loadings are applied. These displacement and force measurements are actually taken to fulfill the minimal requirements for unique identification of the elastic moduli $\boldsymbol{C}$ in a piece-wise homogeneous medium.

To summarize, this paper considers two types of measurements on piece-wise homogeneous anisotropic elastic media; i.e., static displacements $\boldsymbol{u}$ on the entire external and internal boundaries, and nonzero surface forces $\boldsymbol{f}$ on at least a finite portion of the external surface.

C. Principle and Uniqueness

Elastography reconstructs the material distribution $\boldsymbol{C}(\boldsymbol{x})$ using measured displacements $\boldsymbol{u}(\boldsymbol{x})$ and forces $\boldsymbol{f}$ as inputs. It is critically important to investigate whether the measurements carry sufficient constitutive information that gives a unique $\boldsymbol{C}(\boldsymbol{x})$. There have been mathematical theories that address this issue for generally inhomogeneous media, assuming various types of material symmetry and static mechanical measurements. For elastography where the full displacement field $\boldsymbol{u}(\boldsymbol{x})$ is measured [30], [31], it is shown that multiple sets of $\boldsymbol{u}(\boldsymbol{x})$ are necessary to avoid ambiguous reconstruction results for anisotropic $\boldsymbol{C}(\boldsymbol{x})$. There are also theories assuming measurements only on the boundary of an inhomogeneous medium. They take as inputs the so-called Dirichlet-to-Neumann (DtN) map, which maps the mathematical *space* of boundary displacements to the *space* of distributed surface forces. Nakamura and co-workers proved that the Lamé parameters $(\lambda(\boldsymbol{x}), \mu(\boldsymbol{x}))$ in two-dimensional (2-D) heterogeneous isotropic elastic media can be uniquely determined from a static DtN map [32], whereas the material parameters in 2-D anisotropic media *cannot* be uniquely identified [33]. They further proved unique identifiability in inhomogeneous 3-D transversely isotropic materials under some conditions [33]. Theory for more general 3-D anisotropic elastography using static DtN map, however, has not been found in the literature. Research efforts on the dynamic boundary elastography for 3-D inhomogeneous media have been conducted by Mazzucato and Rachele [34], who presented proof for unique identification of isotropic media using time-dependent DtN map and partial uniqueness for anisotropic cases.

The surface measurements described in this paper are different from the DtN map. In fact, complete knowledge of a DtN map needs displacement and force measurements on the entire external boundary with an *infinite* number of experiments, which is not feasible in practice. In the present model, distributed forces are assumed accessible on only a portion of the external surface with a few experiments. The constitutive information carried is, thus, much less than that with a complete DtN map, which itself does not guarantee unique results for a general anisotropic inhomogeneous medium. Therefore, two additional types of information are employed to enhance the uniqueness in this paper, based on the capability of the available experimental techniques. The piece-wise homogeneous assumption narrows the elastic modulus field $\boldsymbol{C}(\boldsymbol{x})$ to N unknown but constant tensors $\boldsymbol{C}^I$ $(I = 1, 2, \ldots, N)$, and, thus, significantly reduces the uncertainty for reconstruction, yet is still representative for some tumor-containing biological objects. The displacements on the internal boundaries provide extra constitutive information. Furthermore, multiple sets of the displacement and force measurements are employed to rule out ambiguous reconstruction results [31]. It is noted that $\boldsymbol{C}^I$ in a piece-wise homogeneous medium may be recovered directly with characteristic wavelengths; for instance, for isotropic [6]–[8] and transversely isotropic [25], [26] media, where the dynamic displacement fields are assumed accurately measured. This paper provides an alternative approach for some clinical applications that favor static measurements on the boundaries.

Therefore, the present elastography study is based on the following principle: *For a linear elastic medium consisting of N homogeneous anisotropic regions, if displacements $\boldsymbol{u}$ on the entire external and internal boundaries, and nonzero surface forces $\boldsymbol{f}$ on at least a finite portion of the external surface are both measured, the elastic moduli for all the N regions, $\boldsymbol{C}^I$ $(I = 1, 2, \ldots, N)$, can be reconstructed from finite sets of the measurements.*

For a homogeneous anisotropic elastic medium with controllable boundary displacements, the proposed technique is easily proved. Given a strain tensor $\overline{\varepsilon}$, linear displacements $\boldsymbol{u} = \overline{\boldsymbol{\varepsilon}} \cdot \boldsymbol{x}$ can be applied on the entire boundary $\partial\Omega$. This necessarily gives uniform strain and stress fields in Ω, i.e., $\boldsymbol{\varepsilon}(\boldsymbol{x}) = \overline{\boldsymbol{\varepsilon}}$ and $\boldsymbol{\sigma}(\boldsymbol{x}) = \overline{\boldsymbol{\sigma}}$, where $\overline{\boldsymbol{\sigma}}$ can be recovered from the distributed forces $\boldsymbol{f}$ measured on at least a finite portion of the external surface, which is not ruled for biological objects. The anisotropic elastic modulus $\boldsymbol{C}$ can be uniquely determined with six linearly independent strains $\overline{\boldsymbol{\varepsilon}}$ and the associated stresses $\overline{\boldsymbol{\sigma}}$.

For those cases where the displacements are controllable only on *part* of the boundary, the identification is not straightforward. Therefore, we give a finite-element-based justification for the proposed principle. Following the standard finite element procedures [35], and assuming no body force, an elastic boundary problem is discretized as

$$[K_{\mathrm{B}}] \left\{ \mathrm{u}_{\mathrm{B}}^{(i)} \right\} = \left\{ f_{\mathrm{B}}^{(i)} \right\} \tag{2}$$

in which $\mathrm{u}_{\mathrm{B}}^{(i)}$ and $f_{\mathrm{B}}^{(i)}$ are nodal displacements, and forces on the boundary for the ith experiment $(i = 1, 2, \ldots, M)$, respectively. They have the same dimension J. The $J \times J$ stiffness matrix $[K_{\mathrm{B}}]$ depends on the geometry and elastic modulus $\boldsymbol{C}$ in Ω. According to the assumed measurements, all components of $\mathrm{u}_{\mathrm{B}}^{(i)}$ and $L^{(i)}$ $(1 \leq L^{(i)} \leq J)$ components of $f_{\mathrm{B}}^{(i)}$ are known. They yield $L^{(i)}$ equations about the unknown $\boldsymbol{C}$ in the forms of $\sum_{s=1,J} [K_{\mathrm{B}}]_{ms} \left\{ \mathrm{u}_{\mathrm{B}}^{(i)} \right\}_s = \left\{ f_{\mathrm{B}}^{(i)} \right\}_m$, where the subscripts denote components. The M experiments give $L_{\mathrm{Total}} = \sum_{i=1,M} L^{(i)}$ equations. With a large enough number of well-designed experiments, it is expected that all the components of $\boldsymbol{C}$ can be solved from these L_{Total} equations.

The above justification can be directly generalized for a piece-wise homogeneous medium. The elastic modulus $\boldsymbol{C}^1$ for the matrix Ω^1 is first identified from measurements of the displacements $\mathrm{u}_{\mathrm{B}}^{(i)}$ on its entire boundary (including $\partial\Omega$ and $\partial\Omega^I$, see

Fig. 1) and part of the surface forces $f_{\mathrm{B}}^{(i)}$ in a finite number of experiments. With the identified $\boldsymbol{C}^1$, $[K_{\mathrm{B}}]$ for the matrix Ω^1 is obtained, and then $f_{\mathrm{B}}^{(i)}$ becomes completely known by (2). Now, for a region completely embedded in Ω^1, say Ω^2, its external boundary is a part of the boundary of Ω^1, so that *both* its surface displacements and forces are attainable, and its elastic modulus is determinable. For a region adjacent to Ω^1, say Ω^3, the surface forces are known on part of its boundary, which also enable identification of $\boldsymbol{C}^3$. By repeating these procedures, the elastic moduli $\boldsymbol{C}^I$ $(I = 1, 2, 3, \ldots, N)$ for all the N constituent regions are reconstructed.

III. Algorithm

While the previous section has explained the feasibility of the proposed principle for anisotropic elastography, it does not provide an efficient reconstruction method, since the expressions for the stiffness matrix $[K_{\mathrm{B}}]$ as functions of $\boldsymbol{C}$ vary with the geometry, boundary conditions and discretization method, and are unknown in general. In this section, an optimization-based algorithm is proposed for reconstructing the anisotropic elastic moduli.

A. Objective Function and Equilibrium Partial Differential Equation (PDE)

An elastography problem can be stated so as to find a distribution of elastic modulus $\boldsymbol{C}(\boldsymbol{x})$, which optimally minimizes the difference between the *measured* displacements $\boldsymbol{U}^m$ and the *calculated* displacements $\boldsymbol{u}$ that depend on $\boldsymbol{C}(\boldsymbol{x})$, i.e., to minimize the error function

$$\Phi\left(\boldsymbol{C}\left(\boldsymbol{x}\right)\right) = \int_{\Omega} \left(\boldsymbol{u} - \boldsymbol{U}^m\right) \cdot \boldsymbol{\chi}_{\Omega}\left(\boldsymbol{x}\right) \cdot \left(\boldsymbol{u} - \boldsymbol{U}^m\right) \mathrm{d}V + \int_{\partial\Omega_t} \left(\boldsymbol{u} - \boldsymbol{U}^m\right) \cdot \boldsymbol{\chi}_{\Gamma}\left(\boldsymbol{x}\right) \cdot \left(\boldsymbol{u} - \boldsymbol{U}^m\right) \mathrm{d}A \quad (3)$$

in which the second-order tensors $\boldsymbol{\chi}_{\Omega}$ and $\boldsymbol{\chi}_{\Gamma}$ are pre-selected weight functions. Note that the assumption of piece-wise homogeneity leads to $\boldsymbol{C}(\boldsymbol{x}) = \boldsymbol{C}^I$ for $\boldsymbol{x} \in \Omega^I$ $(I = 1, 2, \ldots, N)$, and the boundary measurements turn the first integral into a surface integral on the internal boundaries. The calculated displacements $\boldsymbol{u}$ are the solution of the equilibrium PDE

$$\begin{cases} \nabla \cdot \boldsymbol{\sigma}\left(\boldsymbol{u}\right) + \boldsymbol{b} = \boldsymbol{0}, \ \boldsymbol{\sigma}\left(\boldsymbol{u}\right) = \boldsymbol{C}\left(\boldsymbol{x}\right) : \boldsymbol{\varepsilon}\left(\boldsymbol{u}\right) \text{ in } \Omega \\ \boldsymbol{u} = \boldsymbol{U} \text{ on } \partial\Omega_u \text{ and } \boldsymbol{\sigma}\left(\boldsymbol{u}\right) \cdot \boldsymbol{n} = \boldsymbol{f} \text{ on } \partial\Omega_t \end{cases} \quad (4)$$

where $\boldsymbol{b}$ are body forces applied in Ω, and $\boldsymbol{f}$ are distributed forces on the external boundary $\partial\Omega_t$. On the rest of the external boundary, namely $\partial\Omega_u$, displacements $\boldsymbol{u}$ are prescribed as $\boldsymbol{U}$. For a forward elastic problem, either force or displacement component must be prescribed at any point for any degree-of-freedom, without ambiguity. For elastography reconstruction, however, forces and displacements could be *both* known at some portions in Ω or on its boundary. In such a case, the known displacements are treated as measurements $\boldsymbol{U}^m$, and the forces are prescribed, so that (4) is well-posed. Equation (4) is equivalent to a *weak form* that $\boldsymbol{u}$ equal $\boldsymbol{U}$ on $\partial\Omega_u$, and satisfy

$$0 = \int_{\Omega} \boldsymbol{\varepsilon}\left(\boldsymbol{u}\right) : \boldsymbol{C}\left(\boldsymbol{x}\right) : \boldsymbol{\varepsilon}\left(\boldsymbol{w}\right) \mathrm{d}V - \int_{\Omega} \boldsymbol{w} \cdot \boldsymbol{b} \,\mathrm{d}V - \int_{\partial\Omega_t} \boldsymbol{w} \cdot \boldsymbol{f} \mathrm{d}A \quad (5)$$

for arbitrary virtual displacements $\boldsymbol{w}$ that vanish on $\partial\Omega_u$, i.e., $\boldsymbol{w} = \boldsymbol{0}$ on $\partial\Omega_u$. Note that $\boldsymbol{\varepsilon}\left(\boldsymbol{w}\right) = \left(\nabla \boldsymbol{w}\left(\boldsymbol{x}\right) + \left(\nabla \boldsymbol{w}\left(\boldsymbol{x}\right)\right)^T\right)/2$.

B. Adjoint Method for Gradients $\partial\Phi/\partial\boldsymbol{C}$

The simplest optimization methods only require user-supplied objective function $\Phi(\boldsymbol{C}(\boldsymbol{x}))$. The elastic modulus $\boldsymbol{C}(\boldsymbol{x})$ is updated with finite-deference gradients, trial-and-error method or *ad hoc* approximate gradients [10]. Without user-supplied analytic gradients, the optimization-based reconstruction procedure is time-consuming, considering the large number of iterative steps needed for convergent results and the high computational expense for solving (4) in each step. The challenge for analytically calculating the gradients $\partial\Phi/\partial\boldsymbol{C}$ comes from the fact that Φ is a nonlinear implicit function of $\boldsymbol{C}(\boldsymbol{x})$ defined through (3) and (4). Here, analytic formulas are presented for cost-efficient computation of the gradients, with use of an *adjoint* method.

To decouple the implicit relation between $\boldsymbol{C}$ and $\boldsymbol{u}$, the weak-form constraint (5) is introduced into the objective function Φ, and a Lagrangian is formed as

$$\begin{aligned} \mathrm{L}(\boldsymbol{C}, \boldsymbol{u}, \boldsymbol{w}) = & \int_{\Omega} \left(\boldsymbol{u} - \boldsymbol{U}^m\right) \cdot \boldsymbol{\chi}_{\Omega}\left(\boldsymbol{x}\right) \cdot \left(\boldsymbol{u} - \boldsymbol{U}^m\right) \mathrm{d}V \\ & + \int_{\partial\Omega_t} \left(\boldsymbol{u} - \boldsymbol{U}^m\right) \cdot \boldsymbol{\chi}_{\Gamma}\left(\boldsymbol{x}\right) \cdot \left(\boldsymbol{u} - \boldsymbol{U}^m\right) \mathrm{d}A \\ & + \int_{\Omega} \boldsymbol{\varepsilon}(\boldsymbol{u}) : \boldsymbol{C}(\boldsymbol{x}) : \boldsymbol{\varepsilon}(\boldsymbol{w}) \mathrm{d}V - \int_{\Omega} \boldsymbol{w} \cdot \boldsymbol{b} \,\mathrm{d}V - \int_{\partial\Omega_t} \boldsymbol{w} \cdot \boldsymbol{f} \mathrm{d}A \end{aligned} \quad (6)$$

with $\boldsymbol{u} = \boldsymbol{U}$ and $\boldsymbol{w} = \boldsymbol{0}$ on $\partial\Omega_u$. Noting that displacements $\boldsymbol{u}$ satisfy equality (5), it can be shown that

$$\Phi\left(\boldsymbol{C}\right) = \mathrm{L}\left(\boldsymbol{C}, \boldsymbol{u}, \boldsymbol{w}\right) \quad \text{and} \quad \delta\Phi = \delta\mathrm{L} \quad (7)$$

for arbitrary displacement fields $\boldsymbol{w}(\boldsymbol{x})$, $\delta\boldsymbol{w}\left(\boldsymbol{x}\right)$, and $\delta\boldsymbol{u}\left(\boldsymbol{x}\right)$ that equal zero on $\partial\Omega_u$, where "δ" indicates variation. This flexibility allows suitable choice for $\boldsymbol{w}$, with which only simple terms remain in the expression for $\delta\mathrm{L}$ and such that $\delta\Phi$ is computed efficiently. After some variational derivations, the best choice is found that displacements $\boldsymbol{w}$ are defined by the difference between $\boldsymbol{u}$ and $\boldsymbol{U}^m$ through the following equation:

$$\begin{cases} \nabla \cdot \boldsymbol{\sigma}\left(\boldsymbol{w}\right) - 2\left(\boldsymbol{u} - \boldsymbol{U}^m\right) \cdot \boldsymbol{\chi}_{\Omega}\left(\boldsymbol{x}\right) = \boldsymbol{0} \ \boldsymbol{\sigma}\left(\boldsymbol{w}\right) = \boldsymbol{C}\left(\boldsymbol{x}\right) : \boldsymbol{\varepsilon}\left(\boldsymbol{w}\right) \text{ in } \Omega \\ \boldsymbol{w} = \boldsymbol{0} \text{ on } \partial\Omega_u \text{ and } \boldsymbol{\sigma}\left(\boldsymbol{w}\right) \cdot \boldsymbol{n} = -2\left(\boldsymbol{u} - \boldsymbol{U}^m\right) \cdot \boldsymbol{\chi}_{\Gamma}\left(\boldsymbol{x}\right) \text{ on } \partial\Omega_t \end{cases} . \quad (8)$$

With this special choice, the gradients $\partial\Phi/\partial\boldsymbol{C}$ are readily obtained from

$$\delta\Phi = \int_{\Omega} \boldsymbol{\varepsilon}\left(\boldsymbol{u}\right) : \delta\boldsymbol{C}\left(\boldsymbol{x}\right) : \boldsymbol{\varepsilon}\left(\boldsymbol{w}\right) \mathrm{d}V. \quad (9)$$

These formulas reflect the original idea of the continuum-based optimality criteria technique for topology optimization [36], where displacements $\boldsymbol{w}$ defined in (8) are called the *adjoint* displacements. Oberai *et al.* [17] implemented the adjoint method for isotropic elastography, in which sample reconstructions were performed to identify the distribution of shear modulus in 2-D incompressible media from the displacement field measured in one direction. The formulas (3)–(9) for anisotropic elastography are more general than [17], whereas the variational derivations are similar and are not to be detailed. The most favorable advantage of the adjoint method is that it calculates the gradients $\partial\Phi/\partial\boldsymbol{C}$ most efficiently. By comparing (4) and (8), it can be shown that the real displacements $\boldsymbol{u}$ and the adjoint displacements $\boldsymbol{w}$ share the same stiffness matrix [K] and its Cholesky factorization when finite element methods are employed. Because the calculation processes of [K] and its factorization are the most time-consuming tasks in finite element computations, solving (8) for $\boldsymbol{w}$ and then integrating (9) for $\partial\Phi/\partial\boldsymbol{C}$ add just a fraction of time to the solution for $\boldsymbol{u}$ in (4), which is already required in an optimization-based scheme for elastography reconstruction. Recently, Oberai *et al.* [37] gave a detailed evaluation of the numerical efficiency of elastography reconstructions using the adjoint gradients.

C. Multiple Sets of Measurements

Because most inverse problems in anisotropic elasticity [30], [34] are nonunique in nature, one set of the displacement and force measurements, especially those taken only on the boundaries, may not provide sufficient information for unique identification of $\boldsymbol{C}(\boldsymbol{x})$. Barbone and Gokhale [38] recently proposed the feasibility of using multiple displacement fields to reduce the likelihood of nonuniqueness for 2-D isotropic elastography. Mathematical theories for 3-D elastography using DtN-type boundary measurements are given in [39] for isotropic and transversely isotropic materials. However, uniqueness theory on more general 3-D anisotropic materials with incomplete surface measurements has not been found in the literature. Due to the complexity of 3-D anisotropic elasticity, multiple sets of measurements could be inevitable in practice, where measurements taken in a single experiment may provide only limited information on the material properties. Therefore, the above adjoint method is extended for handling multiple sets of the measurements. Considering M experiments carried on an elastic body Ω, quantities in each experiment are denoted with superscript "(i)" ($i = 1, 2, 3 \ldots, M$). The objective function and its gradients are calculated by (10), shown at the bottom of the page. Displacements $\boldsymbol{u}^{(i)}$ and $\boldsymbol{w}^{(i)}$ for the ith experiment are the solutions of (4) and (8) (with all quantities assigned with superscript "(i)"), respectively. Note that the displacement and traction boundaries $\partial\Omega_u^{(i)}$ and $\partial\Omega_t^{(i)}$ can be different for the M experiments, and so are the weight functions.

D. Incompressible Limit

Deformations of many biological tissues approach the incompressible limit, i.e., the volumetric strain $\varepsilon_{kk} = \nabla \cdot \boldsymbol{u}$ is very low compared to the shear strain. For these tissues, some of the ratios between the components of modulus $\boldsymbol{C}$ become very high; for instance, the ratio λ/μ of an isotropic tissue may reach the order of 10^4 or higher. This introduces numerical problems for biomedical simulations and elastography reconstructions. To address this issue, an incompressible version of the above formulas can be derived, following Oberai *et al.* [17] for isotropic case. Specifically, constraints $\nabla \cdot \boldsymbol{u} = 0$ and $\nabla \cdot \boldsymbol{w} = 0$ are imposed on the displacement fields $\boldsymbol{u}$ and $\boldsymbol{w}$, respectively. The stresses in the governing (4) and (8) become $\boldsymbol{\sigma}(\boldsymbol{u}) = \boldsymbol{C}(\boldsymbol{x}) : \varepsilon(\boldsymbol{u}) - p(\boldsymbol{x})\boldsymbol{I}$ and $\boldsymbol{\sigma}(\boldsymbol{w}) = \boldsymbol{C}(\boldsymbol{x}) : \varepsilon(\boldsymbol{w}) - q(\boldsymbol{x})\boldsymbol{I}$, respectively, where $p(\boldsymbol{x})$ is a hydrostatic pressure field in the medium, and $q(\boldsymbol{x})$ is its adjoint field. No further modification is needed in formulas (3)–(10). It is noted that the incompressible modulus $\boldsymbol{C}$ should be defined in a way different from that of compressible cases, as briefly described in Appendix I.

IV. Simulations

Biomechanical research on the mechanical behavior of tumor-containing breasts has been active in the past decade. Wellman *et al.* [3] performed *ex vivo* experiments and proposed nonlinear elastic models for normal and cancerous breast tissues. Tanner *et al.* [27] then carried out finite element simulations using Wellman's material models. Samani *et al.* [40] conducted large-strain finite element computation on a geometric model reconstructed from MR breast images. Elastography for breast tumor detection is referred to [9], [10], [23], and [41]. Plewes *et al.* [9] employed MR measurements of a quasi-static displacement field as input for reconstructing the isotropic moduli in a breast phantom, and assumed piece-wise homogeneity, as proposed in this paper. There are also inclusion-identification type semi-analytical studies on ideal phantoms [20], [21] that give insights into elastography using boundary measurements.

This section presents anisotropic elastography simulations using boundary measurements on a 3-D human breast phantom, demonstrating the validity and utility of the proposed principle and algorithm. Forward finite element computations are first carried on the breast phantom, assuming isotropic and orthotropic elastic moduli, respectively. Displacements on the entire external and internal boundaries and forces on part of the external surface are then extracted from the forward results, and are used as inputs for elastography reconstructions.

A. Three–Dimensional Breast Phantom and Forward Computations

As shown in Fig. 2, a 3-D breast phantom is employed in our simulations. It consists of a 10 cm half-spherical matrix centering at (0, 0, 0), imitating normal breast tissues, and a 1.5 cm spherical hard inclusion at (2.00, 1.75, 2.25), imitating a tumor. In the following text, terms *tissue* and *tumor* are used to denote the matrix and the inclusion, respectively. Since we do not intend to perform *in vivo* study at this stage, the elastic material parameters of the tissue and tumor are *arbitrarily* chosen for the simulations, and their magnitudes are relative.

$$\begin{cases}\Phi(\boldsymbol{C}(\boldsymbol{x})) = \sum_{i=1}^{M}\left(\int_{\Omega}\left(\boldsymbol{u}^{(i)} - \boldsymbol{U}^{m(i)}\right)\cdot\boldsymbol{\chi}_{\Omega}^{(i)}(\boldsymbol{x})\cdot\left(\boldsymbol{u}^{(i)} - \boldsymbol{U}^{m(i)}\right)\mathrm{d}V + \int_{\partial\Omega_t^{(i)}}\left(\boldsymbol{u}^{(i)} - \boldsymbol{U}^{m(i)}\right)\cdot\boldsymbol{\chi}_{\Gamma}^{(i)}(\boldsymbol{x})\cdot\left(\boldsymbol{u}^{(i)} - \boldsymbol{U}^{m(i)}\right)\mathrm{d}A\right),\\ \delta\Phi = \sum_{i=1}^{M}\int_{\Omega}\varepsilon\left(\boldsymbol{u}^{(i)}\right) : \delta\boldsymbol{C}(\boldsymbol{x}) : \varepsilon\left(\boldsymbol{w}^{(i)}\right)\mathrm{d}V\end{cases} \quad (10)$$

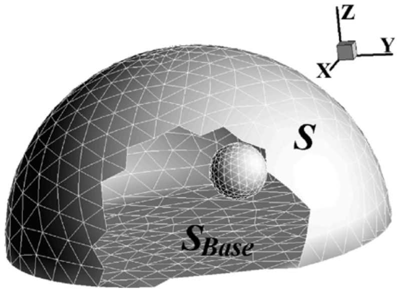

Fig. 2. A 3-D phantom mimicking the normal breast tissues and an embedded tumor. Finite-element mesh is shown on the external and internal boundaries.

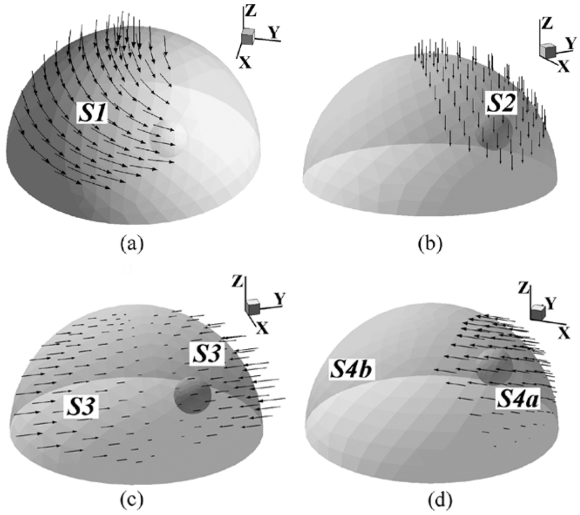

Fig. 3. Four external loading modes applied on the breast phantom. Displacement $\boldsymbol{u} = \mathbf{0}$ on S_{Base} for all loadings, and (a) Loading A, $\boldsymbol{u}$ prescribed on surface $S1$; (b) Loading B, u_z prescribed on $S2$; (c) Loading C, forces F_y prescribed on $S3$; (d) Loading D, u_x prescribed on surface $S4a$, $u_x = u_y = 0$ on $S4b$.

Finite element discretization method [35] is employed for both the forward and reconstruction computations. The phantom is discretized with 1534 nodes, and 1874 and 7497 tetrahedral elements for the tumor and tissue, respectively. The base surface where $\text{z} = 0$ is denoted as S_{Base}, and the rest of the external surface as S. Four loading modes are designed to reflect some characteristics of manual palpation for breast tumor detection. For all cases, displacements $\boldsymbol{u} = \mathbf{0}$ on S_{Base}. Surface displacements combining twinning and compression modes are prescribed on portion $S1$ of S for loading A [Fig. 3(a)]. Fig. 3(b) shows uniform vertical displacements $u_z = -0.5$ cm imposed on $S2$ for loading B. Loading D [Fig. 3(d)] is applied with u_x prescribed on $S4a$ and constraints $u_x = u_y = 0$ on $S4b$. For loading C [Fig. 3(c)], surface forces F_y are applied along y-direction on $S3$. The rest of the external surface without displacements or forces specified is traction-free.

The elastic moduli are first assumed isotropic, with Lamé parameters (λ, μ) equal (25, 7.5) for the tissue and (125, 25) for the tumor. Surface deformations corresponding to the four loadings are similar, but different from those in Fig. 4(a)–(d) for anisotropic cases. Then, orthotropic moduli are assumed for the tissue and tumor. Each has three Eulerian orientation angles $(\varphi_1, \phi, \varphi_2)$ and nine material parameters $\{\hat{C}^0_{11}, \hat{C}^0_{22}, \hat{C}^0_{33}, \hat{C}^0_{44}, \hat{C}^0_{55}, \hat{C}^0_{66}, \hat{C}^0_{23}, \hat{C}^0_{13}, \hat{C}^0_{12}\}$ (details in Appendix II). This choice of orthotropy is general enough for the purpose of biomedical applications, as most tissues belong to the orthotropic class. The proposed principle and numerical algorithms should also work for more general anisotropic materials. In the forward computations, the Eulerian angles are (0, 0, 0) and the material parameters are

$$\{\hat{C}\}_{\text{Tissue}} = \{30, 40, 20, 14, 8, 10, 6, 12, 10\}$$

and

$$\{\hat{C}\}_{\text{Tumor}} = \{200,200,120,30,70,55,6,12,10\} \tag{11}$$

such that the tumor is approximately five times stiffer than the tissue, consistent with experimental observations [2], [3], [15]. Deformed configurations corresponding to the four loadings are illustrated in Fig. 4(a)–(d).

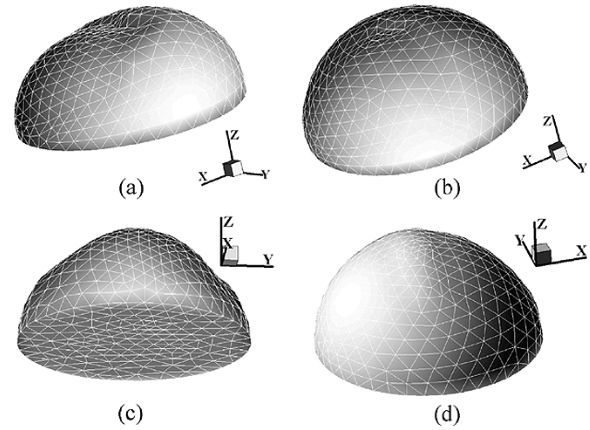

Fig. 4. Surface deformations of breast phantom with orthotropic elastic moduli, corresponding to loading modes (a) A, (b) B, (c) C, and (d) D, respectively.

TABLE I
INPUTS FOR ELASTOGRAPHY RECONSTRUCTIONS

	Prescribed displacements on $\partial\Omega_u$	Prescribed surface forces on $\partial\Omega_t$	Displacement measurements $\boldsymbol{U}^m$
Loading A	$\boldsymbol{u} = \mathbf{0}$ on S_{Base}, u_x & u_y on $S1$	F_z on $S1$, $\boldsymbol{F} = \mathbf{0}$ on $S1^C$	u_z on $S1$, $\boldsymbol{u}$ on $S1^C$ & $\partial\Omega^1$
Loading B	$\boldsymbol{u} = \mathbf{0}$ on S_{Base}	$\boldsymbol{F}$ on S	$\boldsymbol{u}$ on S & $\partial\Omega^1$
Loading C	$\boldsymbol{u} = \mathbf{0}$ on S_{Base}	$\boldsymbol{F}$ on S	$\boldsymbol{u}$ on S & $\partial\Omega^1$
Loading D	$\boldsymbol{u} = \mathbf{0}$ on S_{Base}, $\boldsymbol{u}$ on $S4b$	$\boldsymbol{F}$ on $S4b^C$	$\boldsymbol{u}$ on S & $S4b^C$

B. Reconstructions for Isotropic Phantom

Inputs for the elastography reconstructions are extracted from the forward computational results with loading modes $\text{A} \sim \text{D}$, as detailed in Table I. In the Table, $S1$ and $S4b$ are defined in the previous subsection, and Fig. 3, and $S1^C$ and $S4b^C$ denote the portion of the upper surface S that is not in $S1$ or $S4b$, respectively. At a point where the displacement measurement is taken along the ith coordinate direction, the corresponding weight function χ_i in (3) is set to be 1.0; otherwise $\chi_i = 0$. The reconstruction procedure uses a limited-memory BFGS (L-BFGS)

TABLE II
INITIAL GUESSES & RECONSTRUCTION RESULTS FOR ANISOTROPIC BREAST PHANTOM

	Tissue										Tumor									
	$\hat{C}_{11}$	$\hat{C}_{22}$	$\hat{C}_{33}$	$\hat{C}_{44}$	$\hat{C}_{55}$	$\hat{C}_{66}$	$\hat{C}_{23}$	$\hat{C}_{13}$	$\hat{C}_{12}$	*MaxN*	$\hat{C}_{11}$	$\hat{C}_{22}$	$\hat{C}_{33}$	$\hat{C}_{44}$	$\hat{C}_{55}$	$\hat{C}_{66}$	$\hat{C}_{23}$	$\hat{C}_{13}$	$\hat{C}_{12}$	*MaxN*
Real	30.00	40.00	20.00	14.00	8.00	10.00	6.00	12.00	10.00	0.000	200.0	200.0	120.0	30.00	70.00	55.00	6.00	12.00	10.00	0.000
Inputs	Reconstruction Results																			
A	30.00	40.05	20.00	14.00	7.99	10.00	6.01	12.00	10.04	0.007	**222.8**	115.8	**120.7**	**30.29**	**71.66**	**56.54**	**5.94**	8.47	24.33	2.687
B	30.06	40.14	20.00	13.98	8.00	10.02	5.99	12.02	9.94	0.001	142.2	151.7	**118.7**	38.75	**73.79**	**50.32**	9.66	1.75	41.68	2.874
C	31.16	38.61	19.34	13.92	8.65	9.95	6.15	11.95	10.40	0.454	111.0	**185.4**	154.9	34.01	**68.62**	**50.35**	13.34	8.93	7.09	7.451
D	29.97	39.04	19.98	14.01	8.00	9.98	5.93	11.98	9.84	0.012	**229.4**	80.0	**130.8**	**32.27**	**75.54**	**53.08**	7.28	3.68	5.02	4.673
ABC	30.00	40.00	20.00	14.00	8.00	10.00	6.00	12.00	10.00	0.000	**194.9**	**199.9**	**120.0**	**30.01**	**70.06**	**54.95**	**5.87**	**11.80**	**9.41**	**0.019**
BCD	30.03	40.01	20.00	13.98	7.99	10.03	5.99	12.00	10.02	0.001	150.9	**201.9**	**122.0**	**37.01**	**64.11**	39.00	4.47	**11.48**	**9.40**	1.695
ABCD	30.00	40.00	20.00	14.00	8.00	10.00	6.00	12.00	10.00	0.000	**198.8**	**200.1**	**120.0**	**30.00**	**70.00**	54.96	**5.94**	**11.86**	**9.80**	**0.007**
ABCD -noise	30.17	40.18	20.07	14.31	8.10	10.41	6.12	12.15	10.16	0.323	201.2	201.7	122.4	29.12	65.73	51.96	5.11	16.36	10.37	0.932

optimization method [42]. For a guess of $\boldsymbol{C}(\boldsymbol{x})$, displacements $\boldsymbol{u}$ are solved from (4) and compared with the measurements $\boldsymbol{U}^m$, and the objective function $\Phi(\boldsymbol{C}(\boldsymbol{x}))$ is calculated with (3). Then, the adjoint displacements $\boldsymbol{w}$ and the analytical gradients $\partial\Phi/\partial\boldsymbol{C}$ are obtained via (8) and (9), and are input into a L-BFGS subroutine, where $\boldsymbol{C}(\boldsymbol{x})$ is optimally updated. New iteration is then started until the convergent criterion is met. When the measurements are taken from multiple experiments, formula (10) is used for $\Phi(\boldsymbol{C}(\boldsymbol{x}))$ and $\partial\Phi/\partial\boldsymbol{C}$. In the following simulations, the to-be-identified field of elastic modulus $\boldsymbol{C}(\boldsymbol{x})$ is piece-wise constant, i.e., it is $\{\hat{C}\}_{\text{Tissue}}$ in the tissue and $\{\hat{C}\}_{\text{Tumor}}$ in the tumor.

Lamé parameters (λ, μ) are first identified for the isotropic phantom. With an initial guess (20, 5), the reconstruction with Loading A yields $(25, 7.5)_{\text{Tissue}}$ for the tissue and $(125.05, 25.001)_{\text{Tumor}}$ for the tumor, which are exactly the real material parameters. With other loadings and randomly selected initial guesses, all the reconstructions yield convergent results, with relative errors of less than 0.1%. Reconstructions using multiple sets of the measurements reduce the number of iteration steps. For instance, with inputs from Loadings A, B, and C and initial guess (20, 5), the reconstruction gives $(25, 7.5)_{\text{Tissue}}$ and $(125.04, 25.001)_{\text{Tumor}}$.

C. Reconstructions for Anisotropic Phantom

For the anisotropic phantom, twelve parameters are identified for the tissue and tumor, respectively. They are $(\varphi_1, \phi, \varphi_2)$ and $\{\hat{C}^0_{11}, \hat{C}^0_{22}, \hat{C}^0_{33}, \hat{C}^0_{44}, \hat{C}^0_{55}, \hat{C}^0_{66}, \hat{C}^0_{23}, \hat{C}^0_{13}, \hat{C}^0_{12}\}$. Note that one orthotropic modulus can be represented with different sets of the Eulerian angles and material parameters. For better comparison, $\{\hat{C}_{11}, \hat{C}_{22}, \hat{C}_{33}, \hat{C}_{44}, \hat{C}_{55}, \hat{C}_{66}, \hat{C}_{23}, \hat{C}_{13}, \hat{C}_{12}\}$ are calculated via relations (13)–(15) from the reconstruction results, and compared with the real values at (11), noting that $(\varphi_1, \phi, \varphi_2) = (0, 0, 0)$ for both the tissue and tumor. The other components, $\hat{C}_{mn}$, correspond to zeros in the real elastic moduli. Their maximum value is given as "$\text{Max}N$." The results are given in Table II. Column "Inputs" refers to the loading mode(s) from which the measurements are taken. The guess for Eulerian angels $(\varphi_1, \phi, \varphi_2)$ is randomly set at (1, 0.6, 1.5) (in rad), and that for the material parameters is {20, 20, 20, 5, 5,5, 15, 15, 15}, which describes an isotropic modulus.

Elastography reconstructions are first performed with measurements from a single loading mode. With Loading A, all components for $\{\hat{C}\}_{\text{Tissue}}$ are accurately identified with low relative errors, and are insensitive to the initial guess. For $\{\hat{C}\}_{\text{Tumor}}$ of the embedded tumor, components $\hat{C}_{11}$, $\hat{C}_{33}$, $\hat{C}_{44}$, $\hat{C}_{55}$, $\hat{C}_{66}$, and $\hat{C}_{23}$ (bold-face numbers in Table II) are successfully reconstructed, while the other components do not show convergency even when both $\Phi(\boldsymbol{C}(\boldsymbol{x}))$ and $\partial\Phi/\partial\boldsymbol{C}$ have reached very low values. Other randomly selected initial guesses have been used. It is found that the accurate results (bold-face numbers) are not sensitive to the initial guess. The results with Loadings B, C, and D, individually, are similar; however, the accurately identified components in $\{\hat{C}\}_{\text{Tumor}}$ are different. In fact, measurements from Loadings B and C, individually, only identify three material parameters. This observation of partial convergence may be explained by the dominant deformation modes in response to the applied loadings, as will be discussed.

Reconstructions are further performed using multiple sets of the measurements, i.e., from loadings ABC, BCD, and ABCD, etc. All the components for $\{\hat{C}\}_{\text{Tissue}}$ are accurately and robustly identified. The error eventually disappears with ABC and ABCD combinations, and is within the range of about 0.3% for BCD combination, which is very low compared to the reconstruction errors with individual Loadings B, C, and D, respectively. The tumor, while reconstructions with loadings ABC and BCD are much more accurate and robust than the previous ones, shows error for ABC combination, and leaves $\hat{C}_{11}$, $\hat{C}_{66}$, and $\hat{C}_{23}$ undetermined for BCD. Recognizing that the three-loading combinations may not activate significantly some necessary deformation states for complete identification of $\{\hat{C}\}_{\text{Tumor}}$, reconstructions are further conducted using measurements from all the four loading modes. Elastic parameters for the tumor are now completely identified with improved accuracy. The essential parameters for the tumor, $\hat{C}_{11}$, $\hat{C}_{22}$, $\hat{C}_{33}$, $\hat{C}_{44}$, $\hat{C}_{55}$, and $\hat{C}_{66}$ that make the diagonal terms in the matrix form (12), are precisely identified. Results for the off-diagonal components, $\hat{C}_{23}$, $\hat{C}_{13}$, and $\hat{C}_{12}$, are also satisfactory.

V. Discussion

A. Loading Modes

In the sample anisotropic elastography reconstructions, measurements with a single loading can be adequate for accurate and unique identification of the elastic modulus for the tissue. However, it is insufficient for the tumor. This observation may be explained by the requirements for characterization of the elastic tensor $\boldsymbol{C}$ at a material point, that is six linearly independent sets of $(\boldsymbol{\sigma}, \varepsilon)$ states. With loading modes A, B, and D, movement of the tumor in the z-direction is constrained by the fixed base surface, so that the stress and strain components ε_{33} and σ_{33} are more significant than others. Consequently, $\hat{C}_{33}$ for the tumor can be identified with any one of these loadings and their combinations. On the other hand, deformation of the tumor is less confined in the x- and y-directions, so that determination of $\hat{C}_{11}$ and $\hat{C}_{22}$ depends on the mode of the loading. Loading C emphasizes compression in the y-direction and provides enough information for $\hat{C}_{22}$, but places little constraint on the x- and z-directions, which leads to inaccurate results for $\hat{C}_{11}$ and $\hat{C}_{33}$. The deformation of the tissue is complex with any loading, and provides adequate information for unique identification of the elastic properties. Therefore, using multiple sets of measurements should bring out more constitutive information and smooth out possible errors. Due to the same reason, we expect that the large size, shallow location and complicated shape of the tumor will produce favorable deformations for elastography reconstruction.

The above explanation also serves as a guideline for design of imaging protocols in clinical applications. Reducing the number of necessary loadings increases the clinical efficiency, reduces X-ray dose for CT-based elastography, and benefits the patient. The Loadings A, B, C, and D applied on the breast phantom are picked *ad hoc* and are obviously not optimal. The reconstruction with Loading A furnishes hints for the optimal design. Loading A combines compression and twisting, leads to complex deformation and provides the richest stress-strain information. Consequently, only three parameters for the tumor are left undetermined, compared to at least four with the other loadings individually. Therefore, we expect that at least one and at most three well-designed loadings combining different deformation modes will be sufficient for 3-D anisotropic (orthotropic) elastography with similar anatomic structures. We also expect that realistic elastic phantoms consisting of more, complex piece-wise homogeneous regions may not require more loading modes for complete identification, as the associated deformations will be complex and rich in constitutive information.

B. Perturbation Analysis

The anisotropic elastography has been explored with "ideal" measurements extracted from highly accurate forward computation results, with a relative cut-off error of $< 0.05\%$. Here, we investigate robustness of the proposed method against errors with measurements. In practice, the errors include inevitable noise and artifacts with CT images [43] and mechanical measurements. To simulate these errors, we add noise to the "ideal" displacement and force measurements. The amount of relative error for a nonzero component of the measurements is randomly selected between -5% and 5%, which sufficiently covers the fidelity range of CT image boundary detection, and is also considered typical with *in situ* mechanics measurements using advanced electro-mechanical force sensors.

The results are given as "ABCD-noise" in Table II, showing about 5% maximum relative errors for the elastic parameters of the tissue, and higher for the tumor, compared with the maximum error of 1% with results "ABCD" using ideal inputs. The diagonal terms of the matrix representation (12) of the modulus $\boldsymbol{C}$, $\hat{C}_{11}$, $\hat{C}_{22}$, $\hat{C}_{33}$, $\hat{C}_{44}$, $\hat{C}_{55}$, and $\hat{C}_{66}$, are well identified for the tumor, with a maximum relative error of 6% for $\hat{C}_{55}$ and $\hat{C}_{66}$. Larger disagreements are shown for the off-diagonal terms $\hat{C}_{23}$, $\hat{C}_{13}$, and $\hat{C}_{12}$. This could be explained by the roles of the elastic parameters in the stress-strain states associated with the loading modes, as have been discussed above. Noting the fact that the "ideal" measurements also carry small errors, we expected that, if the cut-off error is reduced significantly, some of those "undeterminable" components in Table II may become identifiable. Multiple sets of the measurements suppress the errors effectively and reveal some weak stress-strain relations that may be invisible with only a single set.

C. Systematic Evaluation

While our simulations have demonstrated the feasibility of the proposed 3-D anisotropic elastography and provided valuable guidelines, a systematic evaluation study is deemed necessary. Specifically, the method should be thoroughly tested for accuracy, reproducibility and robustness. Experimental investigations may include not only measurement errors but also variability of underlying structures. Our initial results using imperfect inputs have been promising, but future work will be needed to completely understand the effects of different error sources, including those in displacements and forces [19]. While the current tomographic techniques ensure highly precise measurements in 3-D, the boundaries of internal structures can be fuzzy, subjective and prone to human error, especially in cases where the material properties do not exhibit sharp contrast on the interface. Therefore, optimal segmentation and registration algorithms must be developed in specific applications and must be evaluated and validated in a category-by-category fashion.

Contrast among the elastic moduli will be critically important for image quality. The "contrast" refers to the ratio between the elastic moduli of the hard parts and the soft parts in an elastic medium. In the given numerical examples, the contrast is about five, i.e., modulus for the tumor is about five times the surrounding tissue. Our numerical experiments indicate that the reconstruction accuracy not only depends on the stress-strain information supplied and the accuracy of the measurements, but also depends on the contrast. Noting that the stiffness serves as a key indicator that distinguishes benign tumors from malignant ones [44], the assumption of a good contrast mechanism is, therefore, clinically well motivated.

There is another type of "contrast" that plays an important role in biomedical elastography, i.e., the nearly incompressible deformations of many tissues, or the high ratio between shear and volumetric strains. For some isotropic tissues, the ratio λ/μ could be as high as 10^4, compared with the currently used 4–5. In these cases, the parameters related to the bulk stress-strain relation (λ for isotropic materials) are too high to identify accurately, and may affect reconstruction of the other parameters. As a standard treatment for nearly incompressible deformation, the materials can be described as perfectly incompressible, and only the parameters related to the shear stress-strain relation are reconstructed, using the incompressible version of the proposed formulas (Section III-D). The influences of incompressibility in elastography has been discussed recently for isotropic materials [17], and needs further investigation for general anisotropic materials in the future.

VI. Conclusion

A governing principle and a corresponding algorithm have been established for anisotropic elastography. A tumor-containing heterogeneous biomedical tissue has been modeled as being piece-wise homogeneous with anisotropic elastic moduli. The proposed principle for identification of the elastic moduli incorporates both displacement and force measurements on the boundaries. The displacements are measured on all the internal and external boundaries of the medium, with independent biomedical tomography modalities. The necessary surface forces are obtained on a portion of the external boundary. An optimization-based practical algorithm has been developed for anisotropic elastography using these boundary measurements. Particularly, the algorithm allows use of multiple sets of the measurements, which is often important for achieving a unique and stable solution.

The principle has been applied for identification of the anisotropic elastic moduli in a 3-D breast phantom consisting of soft normal tissue and an embedded hard tumor. The simulation results support the proposed framework and techniques for anisotropic elastography of piece-wise homogeneous, anisotropic biomedical objects; i.e., the elastic moduli of the constituent parts can be reconstructed from finite sets of the measurements of the boundary displacements and part of the distributed surface forces. It has been found that while the isotropic moduli can be accurately identified from measurements taken with one loading, 3-D anisotropic elastography typically needs multiple measurements for a unique and accurate solution. This paper opens the door toward anisotropic elastography, provides insight into tomography-based elastography in general, and gives guidelines for future developments and clinical applications such as cancer detection and staging.

Appendix I
General Anisotropic Elastic Materials

The elastic constitutive relation $\boldsymbol{\sigma} = \boldsymbol{C} : \boldsymbol{\varepsilon}$ in 3-D space can be represented with a matrix form in six-dimensional space [45], i.e., $\hat{\boldsymbol{\sigma}} = \hat{\boldsymbol{C}} \cdot \hat{\boldsymbol{\varepsilon}}$, where

$$\hat{\boldsymbol{\sigma}} = \begin{Bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sqrt{2}\sigma_{23} \\ \sqrt{2}\sigma_{13} \\ \sqrt{2}\sigma_{12} \end{Bmatrix}, \quad \hat{\boldsymbol{\varepsilon}} = \begin{Bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ \sqrt{2}\varepsilon_{23} \\ \sqrt{2}\varepsilon_{13} \\ \sqrt{2}\varepsilon_{12} \end{Bmatrix}$$

and

$$\hat{\boldsymbol{C}} = \begin{pmatrix} C_{1111} & C_{1122} & C_{1133} & \sqrt{2}C_{1123} & \sqrt{2}C_{1113} & \sqrt{2}C_{1112} \\ & C_{2222} & C_{2233} & \sqrt{2}C_{2223} & \sqrt{2}C_{2213} & \sqrt{2}C_{2212} \\ & & C_{3333} & \sqrt{2}C_{3323} & \sqrt{2}C_{3313} & \sqrt{2}C_{3312} \\ & \text{sym.} & & 2C_{2323} & 2C_{2313} & 2C_{2312} \\ & & & & 2C_{1313} & 2C_{1312} \\ & & & & & 2C_{1212} \end{pmatrix}. \quad (12)$$

The constitutive relation can be alternatively represented as $\boldsymbol{\varepsilon} = \boldsymbol{M} : \boldsymbol{\sigma}$, where $\boldsymbol{M} = \boldsymbol{C}^{-1}$ is the compliance tensor. Incompressible materials are defined with constraints $M_{ijkk} = 0$ for any set of ij [46]. Similar to (12), the relation between the stress deviator $\boldsymbol{s}$ ($= \boldsymbol{\sigma} - \boldsymbol{I}\sigma_{kk}/3$) and the trace-free strain $\boldsymbol{\varepsilon}$ ($\varepsilon_{kk} = 0$) in incompressible materials can be represented with a matrix form in five-dimensional space, i.e., $\breve{\boldsymbol{s}} = \breve{\boldsymbol{C}} \cdot \breve{\boldsymbol{\varepsilon}}$. The maximum number of independent components in $\breve{\boldsymbol{C}}$ is 15, compared with 21 for compressible materials.

Appendix II
Orthotropic Materials

Description of the stiffness matrix $\hat{\boldsymbol{C}}$ for orthotropic materials needs 3 Eulerian angles $\{\varphi_1, \phi, \varphi_2\}$ and 9 material parameters $\{\hat{C}_{11}^0, \hat{C}_{22}^0, \hat{C}_{33}^0, \hat{C}_{44}^0, \hat{C}_{55}^0, \hat{C}_{66}^0, \hat{C}_{23}^0, \hat{C}_{13}^0, \hat{C}_{12}^0\}$ (e.g., [46])

$$\hat{\boldsymbol{C}} = \boldsymbol{Q} \begin{pmatrix} \hat{C}_{11}^0 & \hat{C}_{12}^0 & \hat{C}_{13}^0 & 0 & 0 & 0 \\ & \hat{C}_{22}^0 & \hat{C}_{23}^0 & 0 & 0 & 0 \\ & & \hat{C}_{33}^0 & 0 & 0 & 0 \\ & \text{sym.} & & \hat{C}_{44}^0 & 0 & 0 \\ & & & & \hat{C}_{55}^0 & 0 \\ & & & & & \hat{C}_{66}^0 \end{pmatrix} \boldsymbol{Q}^T \quad (13)$$

in which the orientation matrix $\boldsymbol{Q}$ defines the principle axes of symmetry; it is a 6×6 matrix, shown in (14) at the bottom of the page, where R_{mn} are components of a second-order orientation

$$\boldsymbol{Q} = \begin{pmatrix} R_{11}^2 & R_{12}^2 & R_{13}^2 & \sqrt{2}R_{12}R_{13} & \sqrt{2}R_{11}R_{13} & \sqrt{2}R_{11}R_{12} \\ R_{21}^2 & R_{22}^2 & R_{23}^2 & \sqrt{2}R_{22}R_{23} & \sqrt{2}R_{21}R_{23} & \sqrt{2}R_{22}R_{21} \\ R_{31}^2 & R_{32}^2 & R_{33}^2 & \sqrt{2}R_{33}R_{32} & \sqrt{2}R_{33}R_{31} & \sqrt{2}R_{31}R_{32} \\ \sqrt{2}R_{21}R_{31} & \sqrt{2}R_{22}R_{32} & \sqrt{2}R_{23}R_{33} & R_{12}R_{13}+R_{12}R_{13} & R_{21}R_{33}+R_{31}R_{23} & R_{21}R_{32}+R_{31}R_{22} \\ \sqrt{2}R_{11}R_{31} & \sqrt{2}R_{12}R_{32} & \sqrt{2}R_{13}R_{33} & R_{12}R_{33}+R_{32}R_{13} & R_{11}R_{33}+R_{13}R_{31} & R_{11}R_{32}+R_{31}R_{12} \\ \sqrt{2}R_{11}R_{21} & \sqrt{2}R_{12}R_{22} & \sqrt{2}R_{13}R_{23} & R_{12}R_{23}+R_{22}R_{13} & R_{11}R_{23}+R_{21}R_{13} & R_{11}R_{22}+R_{21}R_{12} \end{pmatrix} \quad (14)$$

$$\boldsymbol{R} = \begin{pmatrix} C\varphi_1 C\varphi_2 - S\varphi_1 S\varphi_2 C\phi & S\varphi_1 C\varphi_2 + C\varphi_1 S\varphi_2 C\phi & S\varphi_2 S\phi \\ -C\varphi_1 S\varphi_2 - S\varphi_1 C\varphi_2 C\phi & -S\varphi_1 S\varphi_2 + C\varphi_1 C\varphi_2 C\phi & C\varphi_2 S\phi \\ S\varphi_1 S\phi & -C\varphi_1 S\phi & C\phi \end{pmatrix}, \begin{pmatrix} C\varphi_1 = \cos\varphi_1 & S\varphi_1 = \sin\varphi_1 \\ C\phi = \cos\phi & S\phi = \sin\phi \\ C\varphi_2 = \cos\varphi_2 & S\varphi_2 = \sin\varphi_2 \end{pmatrix} \tag{15}$$

$$\begin{cases} A_{16} = -A_{26} = \sqrt{2}, \ A_{45} = -1; \\ B_{15} = -B_{35} = \sqrt{2}S\varphi_2, \ B_{24} = -B_{34} = \sqrt{2}C\varphi_2, \ B_{46} = -S\varphi_2, \ B_{56} = -C\varphi_2 \end{cases} \tag{17}$$

tensor $\boldsymbol{R}$, whose matrix form is shown in (15) at the top of the page. Bunge's convention for the Eulerian angles [46] has been used.

Gradients of the stiffness matrix $\hat{\boldsymbol{C}}$ with respect to the angles can be obtained by

$$\frac{\partial \hat{\boldsymbol{C}}}{\partial \varphi_1} = \boldsymbol{Q}\left(\boldsymbol{A}\hat{\boldsymbol{C}}_0 - \hat{\boldsymbol{C}}_0\boldsymbol{A}\right)\boldsymbol{Q}^T, \quad \frac{\partial \hat{\boldsymbol{C}}}{\partial \phi} = +\boldsymbol{B}\hat{\boldsymbol{C}} - \hat{\boldsymbol{C}}\boldsymbol{B}$$

and

$$\frac{\partial \hat{\boldsymbol{C}}}{\partial \varphi_2} = \boldsymbol{A}\hat{\boldsymbol{C}} - \hat{\boldsymbol{C}}\boldsymbol{A} \tag{16}$$

where $\boldsymbol{A}$ and $\boldsymbol{B}$ are both sparse and anti-symmetric 6×6 matrices, i.e., $A_{nm} = -A_{mn}$ and $B_{nm} = -B_{mn}$, with the nonzero components in the upper triangles shown in (17) at the top of the page. The incompressible version of these formulas are referred to in [46].

ACKNOWLEDGMENT

The authors would like to thank Prof. W. Han of the University of Iowa for helpful discussion. They would also like to thank the reviewers for their thoughtful comments.

REFERENCES

[1] C. P. McPherson, K. K. Swenson, G. Jolitz, and C. L. Murray, "Survival of women ages 40–49 years with breast carcinoma according to method of detection," *Cancer*, vol. 79, pp. 1923–1932, 1997.

[2] A. P. Sarvazyan, A. R. Skovaroda, S. Y. Emelianov, J. B. Fowlkes, J. G. Pipe, R. S. Adler, R. B. Buxton, and P. L. Carson, "Biophysical bases of elasticity imaging," *Acoust. Imag.*, vol. 21, pp. 223–240, 1995.

[3] P. Wellman, R. Howe, E. Dalton, and K. A. Kern, "Breast Tissue Stiffness in Compression is Correlated to Histological Diagnosis: Harvard BioRobotics Laboratory Tech. Rep.," Harvard Univ., Cambridge, MA, 1999.

[4] J. Ophir, I. Cespedes, H. Ponnekanti, Y. Yazdi, and X. Li, "Elastography – A quantitative method for imaging the elasticity of biological tissues," *Ultrason. Imag.*, vol. 13, pp. 111–134, 1991.

[5] L. Gao, K. J. Parker, R. M. Lerner, and S. F. Levinson, "Imaging of the elastic properties of tissue – A review," *Ultrasound Med. Biol.*, vol. 22, pp. 959–977, 1996.

[6] R. Muthupillai, D. J. Lomas, P. J. Rossman, J. F. Greenleaf, A. Manduca, and R. L. Ehman, "Magnetic-resonance elastography by direct visualization of propagating acoustic strain waves," *Science*, vol. 269, pp. 1854–1857, 1995.

[7] A. Manduca, T. E. Oliphant, M. A. Dresner, J. L. Mahowald, S. A. Kruse, E. Amromin, J. P. Felmlee, J. F. Greenleaf, and R. L. Ehman, "Magnetic resonance elastography: Non-invasive mapping of tissue elasticity," *Med. Image Anal.*, vol. 5, pp. 237–254, 2001.

[8] P. J. McCracken, A. Manduca, J. Felmlee, and R. L. Ehman, "Mechanical transient-based magnetic resonance elastography," *Magn. Reson. Med.*, vol. 53, pp. 628–639, 2005.

[9] D. B. Plewes, J. Bishop, A. Samani, and J. Sciarretta, "Visualization and quantification of breast cancer biomechanical properties with magnetic resonance elastography," *Phys. Med. Biol.*, vol. 45, pp. 1591–1610, 2000.

[10] A. Samani, J. Bishop, and D. B. Plewes, "A constrained modulus reconstruction technique for breast cancer assessment," *IEEE Trans. Med. Imag.*, vol. 20, no. 9, pp. 877–885, Sep. 2001.

[11] J. B. Weaver, E. E. W. Van Houten, M. I. Miga, F. E. Kennedy, and K. D. Paulsen, "Magnetic resonance elastography using 3-D gradient echo measurements of steady-state motion," *Med. Phys.*, vol. 28, pp. 1620–1628, 2001.

[12] F. Kallel and M. Bertrand, "Tissue elasticity reconstruction using linear perturbation method," *IEEE Trans. Med. Imag.*, vol. 15, no. 3, pp. 299–313, Jun. 1996.

[13] J. Ophir, S. K. Alam, B. Garra, F. Kallel, E. Konofagou, T. Krouskop, and T. Varghese, "Elastography: Ultrasonic estimation and imaging of the elastic properties of tissues," *J. Eng. Manage. (Proc. Institution of Mechanical Engineers Part H)*, vol. 213, pp. 203–233, 1999.

[14] J. Ophir, F. Kallel, T. Varghese, M. Bertrand, I. Cespedes, and H. Ponnekanti, "Elastography: A systems approach," *Int. J. Imag. Syst. Technol.*, vol. 8, pp. 89–103, 1997.

[15] A. R. Skovoroda, S. Y. Emelianov, M. A. Lubinski, A. P. Sarvazyan, and M. Odonnell, "Theoretical-analysis and verification of ultrasound displacement and strain imaging," *IEEE Trans. Ultrason., Ferroelect. Freq. Control*, vol. 41, no. 3, pp. 302–313, May 1994.

[16] L. S. Taylor, B. C. Porter, D. J. Rubens, and K. J. Parker, "Three-dimensional sonoelastography: Principles and practices," *Phys. Med. Biol.*, vol. 45, pp. 1477–1494, 2000.

[17] A. A. Oberai, N. H. Gokhale, and G. R. Feijoo, "Solution of inverse problems in elasticity imaging using the adjoint method," *Inverse Prob.*, vol. 19, pp. 297–313, 2003.

[18] S. J. Cox and M. Gockenbach, "Recovering planar Lame moduli from a single-traction experiment," *Math. Mech. Solids*, vol. 2, pp. 297–306, 1997.

[19] Y. N. Zhu, T. J. Hall, and J. F. Jiang, "A finite-element approach for Young's modulus reconstruction," *IEEE Trans. Med. Imag.*, vol. 22, no. 7, pp. 890–901, Jul. 2003.

[20] H. B. Kang, E. Kim, and J. Y. Lee, "Identification of elastic inclusions and elastic moment tensors by boundary measurements," *Inverse Prob.*, vol. 19, pp. 703–724, 2003.

[21] H. T. Liu, L. Z. Sun, G. Wang, and M. W. Vannier, "Analytic modeling of breast elastography," *Med. Phys.*, vol. 30, pp. 2340–2349, 2003.

[22] Y. C. Fung, *Biomechanics: Mechanical Properties of Living Tissues*. New York: Springer-Verlag, 1993.

[23] R. Sinkus, J. Lorenzen, D. Schrader, M. Lorenzen, M. Dargatz, and D. Holz, "High-resolution tensor MR elastography for breast tumor detection," *Phys. Med. Biol.*, vol. 45, pp. 1649–1664, 2000.

[24] J. Weaver, M. Doyley, E. Van Houten, M. Hood, X. C. Qin, F. Kennedy, S. Poplack, and K. Paulsen, "Evidence of the anisotropic nature of the mechanical properties of breast tissue," *Med. Phys.*, vol. 29, pp. 1291–1291, 2002.

[25] J. L. Gennisson, S. Catheline, S. Chaffai, and M. Fink, "Transient elastography in anisotropic medium: Application to the measurement of slow and fast shear wave speeds in muscles," *J. Acoust. Soc. Am.*, vol. 114, pp. 536–541, 2003.

[26] R. Sinkus, M. Tanter, S. Catheline, J. Lorenzen, C. Kuhl, E. Sondermann, and M. Fink, "Imaging anisotropic and viscous properties of breast tissue by magnetic resonance-elastography," *Magn. Reson. Med.*, vol. 53, pp. 372–387, 2005.

[27] C. Tanner, A. Degenhard, J. A. Schnabel, C. Hayes, L. I. Sonoda, M. O. Leach, D. R. Hose, D. L. G. Hill, and D. J. Hawkes, "A comparison of biomechanical breast models: A case study," *Proc. SPIE*, vol. 4684, pp. 1807–1818, 2002.

[28] Z. Wang, Y. Liu, L. Z. Sun, and G. Wang, "Elasto-Mammography: Elastic property reconstruction in breast tissues," presented at the MRS Spring Meeting, San Francisco, CA, 2005.

[29] J. V. Hajnal, D. L. G. Hill, and D. J. Hawkes, *Medical Image Registration*. Boca Raton, FL: CRC, 2001.

[30] P. E. Barbone and J. C. Bamber, "Quantitative elasticity imaging: What can and cannot be inferred from strain images," *Phys. Med. Biol.*, vol. 47, pp. 2147–2164, 2002.

[31] J. R. McLaughlin and J. R. Yoon, "Unique identifiability of elastic parameters from time-dependent interior displacement measurement," *Inverse Prob.*, vol. 20, pp. 25–45, 2004.

[32] G. Nakamura and G. Uhlmann, "Inverse problems at the boundary for an elastic medium," *SIAM J. Math. Anal.*, vol. 26, pp. 263–279, 1995.

[33] G. Nakamura, K. Tanuma, and G. Uhlmann, "Layer stripping for transversely isotropic elastic medium," *SIAM J. Appl. Math.*, vol. 59, pp. 1879–1891, 1999.

[34] A. L. Mazzucato and L. V. Rachele, "Partial Uniqueness and Obstruction to Uniqueness in Inverse Problems for Anisotropic Elastic Media," Mathematical Sciences Research Institute (MSRI), Preprints #2003–009 2003.

[35] H. Kardestuncer and D. H. Norrie, *Finite Element Handbook*. New York: McGraw-Hill, 1987.

[36] G. I. N. Rozvany, M. P. BendsØe, and U. Kirsh, "Layout optimization of structures," *Appl. Mech. Rev.*, vol. 48, pp. 41–119, 1995.

[37] A. A. Oberai, N. H. Gokhale, M. M. Doyley, and J. C. Bamber, "Evaluation of the adjoint equation based algorithm for elasticity imaging," *Phys. Med. Biol.*, vol. 49, pp. 2955–2974, 2004.

[38] P. E. Barbone and N. H. Gokhale, "Elastic modulus imaging: On the uniqueness and nonuniqueness of the elastography inverse problem in two dimensions," *Inverse Prob.*, vol. 20, pp. 283–296, 2004.

[39] G. Nakamura and K. Tanuma, "A nonuniqueness theorem for an inverse boundary value problem in elasticity," *SIAM J. Appl. Math.*, vol. 56, pp. 602–610, 1996.

[40] A. Samani, J. Bishop, M. J. Yaffe, and D. B. Plewes, "Biomechanical 3-D finite element modeling of the human breast using MRI data," *IEEE Trans. Med. Imag.*, vol. 20, no. 4, pp. 271–279, Apr. 2001.

[41] A. L. McKnight, J. L. Kugel, P. J. Rossman, A. Manduca, L. C. Hartmann, and R. L. Ehman, "MR elastography of breast cancer: Preliminary results," *Am. J. Roentgenol.*, vol. 178, 2002.

[42] D. C. Liu and J. Nocedal, "On the limited memory BFGS method for large-scale optimization," *Math. Programming*, vol. 45, pp. 503–528, 1989.

[43] J. F. Meinel, G. Wang, M. Jiang, T. Frei, M. Vannier, and E. Hoffman, "Spatial variation of resolution and noise in multi-detector row spiral CT," *Academic Radiol.*, vol. 10, pp. 607–613, 2003.

[44] A. R. Skovoroda, S. Y. Emelianov, and M. Odonnell, "Tissue elasticity reconstruction based on ultrasonic displacement and strain images," *IEEE Trans. Ultrason., Ferroelect., Freq. Control*, vol. 42, no. 4, pp. 747–765, Jul. 1995.

[45] L. J. Walpole, "4th-rank tensors of the 32 crystal classes – Multiplication tables," in *Proc. Roy Soc. Lond. Ser. A-Mathematical Physical and Engineering Sciences*, vol. 391, 1984, pp. 149–179.

[46] Y. Liu, "Macroscopic Behavior, Field Fluctuations and Texture Evolution in Viscoplastic Polycrystals," Ph.D. dissertation, Univ. Pennsylvania, Philadelphia, 2003.